

FitBuddy - AI Fitness Plan Generator

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Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	Project Documentation

Project Description:

FitBuddy is an AI-based web application for generating a personalized seven-day workout plan from a user profile. It collects name, user ID, age, weight, fitness goal and workout intensity, then uses Gemini models to create the workout schedule and a concise nutrition/recovery tip. The user can submit feedback and receive an updated plan. SQLite stores user and plan information for later viewing in the admin page.

The project combines a Jinja2/HTML/CSS frontend, FastAPI backend, Google GenAI integration, and SQLAlchemy-backed SQLite storage. It is designed as an educational prototype demonstrating practical Generative AI integration in a web workflow.

Scenarios

Scenario 1: A user enters age, weight, a muscle-gain goal and medium intensity. FitBuddy generates a seven-day workout schedule with warm-ups, exercises, rest guidance and cooldown/recovery notes.

Scenario 2: A user chooses a weight-loss goal and low intensity. The system generates a gentler weekly routine plus a nutrition/recovery tip focused on sustainable habits.

Scenario 3: A user reviews the plan and submits feedback such as needing lower-impact sessions. The system sends the original plan and feedback to the update generator and returns a revised seven-day plan.

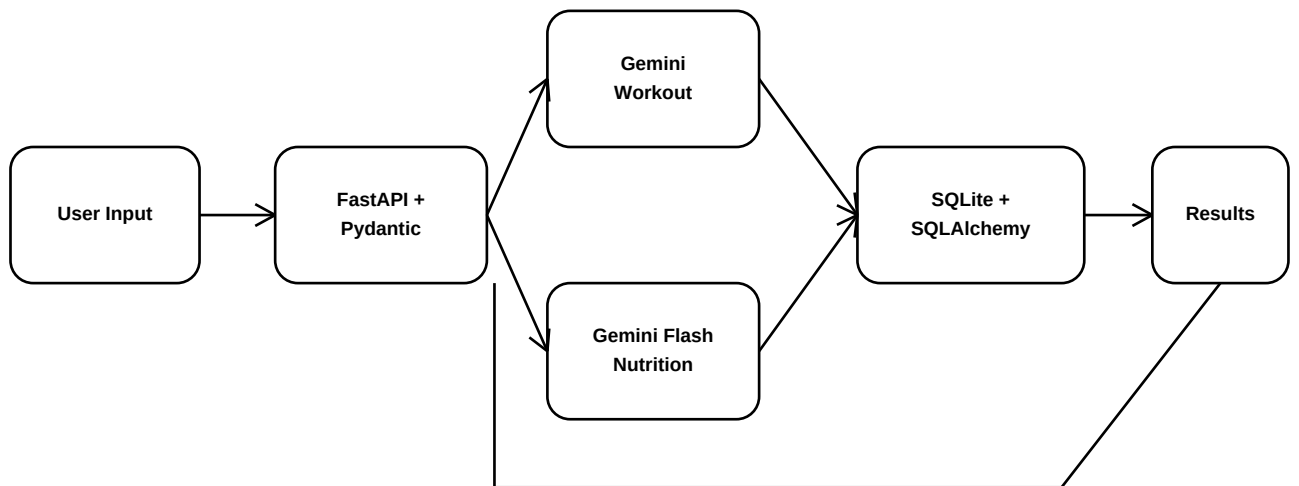
Objectives

- Build a web-based AI fitness planning application.
- Generate a personalized seven-day workout plan.
- Provide nutrition/recovery guidance.
- Enable feedback-based plan updates.
- Store users and plans in SQLite.
- Demonstrate a clean FastAPI + Generative AI architecture.

Technologies Used

Python, FastAPI, Gemini / Google GenAI, HTML, CSS, Jinja2, SQLite, SQLAlchemy, Pydantic, Uvicorn, Pytest, HTTPX and python-dotenv.

System Architecture



Feedback is submitted from Results and sent back to the AI update module.

Project Workflow

User Input -> Validation -> Gemini Workout Generation -> Gemini Flash Nutrition Tip -> Save User/Plan -> Result Display -> Feedback -> Updated Plan -> Save Updated Plan.

Project Modules

main.py starts FastAPI, serves static files/templates and includes routes.
config.py loads environment configuration.
database.py defines User and Plan models plus storage helpers.
schemas.py validates profile inputs.
routes.py connects the web form, AI modules, database and result pages.
gemini_generator.py generates the main workout plan.
gemini_flash_generator.py generates a short nutrition/recovery tip.
updated_plan.py revises the plan from feedback.

Testing

Automated smoke tests are provided for the home page and health endpoint. The project also includes a manual/demo path for AI generation and feedback. No external-service performance number is invented; Gemini latency varies with network and service conditions.

Advantages

- Personalized AI-generated content.
- Simple web interface.
- Seven-day planning.
- Feedback-based revision.
- Database-backed storage.
- FastAPI API documentation.

Limitations

- Current version has no authentication.
- Live AI generation needs internet access and a Gemini API key.
- SQLite is mainly suited to small-scale use.
- Fitness output is informational and not a medical assessment.

Future Enhancements

User authentication; cloud database; progress tracking; workout history; exercise demonstration videos; mobile app; voice interaction; richer AI personalization; cloud deployment; enhanced admin controls; fitness-device integration.

Conclusion

FitBuddy demonstrates a complete Generative AI application flow in which user inputs are validated, sent to Gemini, stored with SQLAlchemy/SQLite and displayed through a responsive web interface. The feedback loop extends the system from one-time generation to interactive revision.

Project Team

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Repository and Demo

Repository: <https://github.com/simson26/TN-Skills>

Demo: <https://drive.google.com/file/d/1tvasdSJQP6KfQRbYd4pb-UOmAHYOTqH/view?usp=drivesdk>